THE UK UNIVERSITY INTEGRATION BEE 2024/2025

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ROUND THREE DICHOTOMY ROUND GRADERS' COPY

Sponsored by



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1 Segment 1

1. (a) $\displaystyle\int_0^2 \sqrt{4-x^2}\,\mathrm{d}x$

   Answer: π

   (b) $\displaystyle\int_0^{\sqrt{2}} \sqrt{4-x^2}\,\mathrm{d}x$

   Answer: $\dfrac{\pi}{2}+1$

2. (a) $\displaystyle\int \frac{\tan(2x)}{\tan(x)}\,\mathrm{d}x$

   Answer: $\dfrac{1}{2}\left(\ln|1+\tan(x)|-\ln|1-\tan(x)|\right)+x+C$

   Answer: $x-\dfrac{1}{2}\left(\ln|\cos(x)-\sin(x)|-\ln|\cos(x)+\sin(x)|\right)$

   Answer: $\tanh^{-1}(\tan(x))+x+C$

   (b) $\displaystyle\int \frac{\tan(x)}{\tan(2x)}\,\mathrm{d}x$

   Answer: $x-\dfrac{\tan(x)}{2}+C$

3. (a) $\displaystyle\int_0^{\infty} \frac{1}{1+e^{\sqrt{x}}}\,\mathrm{d}x$

   Answer: $\dfrac{\pi^2}{6}$

   (b) $\displaystyle\int_0^{\infty} \frac{1}{1+(\sqrt{x})^e}\,\mathrm{d}x$

   Answer: $\dfrac{2\pi}{e}\csc\left(\dfrac{2\pi}{e}\right)$

4. (a) $\displaystyle\int_0^{\infty} \sqrt{x^2\sqrt{2^{-x}\sqrt{x^4\sqrt{4^{-x}\sqrt{x^8\sqrt{8^{-x\sqrt{\cdots}}}}}}}}\,\mathrm{d}x$

   Answer: $\dfrac{729}{32(\ln(2))^3}$

   (b) $\displaystyle\int_0^{\frac{1}{2}} \frac{x}{\sqrt{x+\frac{x}{\sqrt{x+\ldots}}}}+\sqrt{x^3+x^2\sqrt{x^3+x^2\sqrt{x^3+\ldots}}}\,\mathrm{d}x$

   Answer: $\dfrac{1}{4}$

5. (a) $\displaystyle\int_{e^2}^{e^e} \frac{\ln\left(\frac{x^{(\ln(\ln(x)))^2}}{\ln(x)}\right)}{(\ln(\ln(x)))^3\ln(x)}\,\mathrm{d}x$

   Answer: $e^e-\dfrac{e^2}{\ln(2)}$

   (b) $\displaystyle\int_0^1 \ln\left(x^{\ln(1-\ln(x))+1-\ln(x)}\right)\,\mathrm{d}x$

   Answer: -4

6. (a) $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\tan(\sin^2(x))}{1-\tan(\sin^2(x))\tan(\cos^2(x))}\,\mathrm{d}x$

   Answer: $\dfrac{\pi}{4}\tan(1)$

   (b) $\displaystyle\int_0^{\frac{\pi}{4}} \ln\left(\frac{\cos(x)+\sin(x)}{\sqrt{2}\sin(x)}\right)+\ln(\tan(x))\,\mathrm{d}x$

   Answer: 0

7. (a) $\displaystyle\int_0^{1} \frac{1}{(1+x^2)(1+\frac{\pi}{2}+\tan^{-1}(x)\cot^{-1}(x))}\,\mathrm{d}x$

   Answer: $\dfrac{1}{2+\frac{\pi}{2}}\ln\left(1+\dfrac{\pi}{2}\right)$

   (b) $\displaystyle\int_0^{1} \frac{1}{(1+x^2)(1+\tan^{-1}(x)\cot^{-1}(x))}\,\mathrm{d}x$

   Answer: $\dfrac{4}{\sqrt{16+\pi^2}}\tanh^{-1}\left(\dfrac{\pi}{\sqrt{16+\pi^2}}\right)$

   Answer: $\dfrac{2}{\sqrt{16+\pi^2}}\ln\left(\dfrac{\sqrt{16+\pi^2}+\pi}{\sqrt{16+\pi^2}-\pi}\right)$

2 Segment 2

1. (a) $\displaystyle\int_0^1 \frac{\ln(x^4+1)}{x^3}\,\mathrm{d}x$

   Answer: $\dfrac{\pi}{4} - \dfrac{\ln(2)}{2}$

   (b) $\displaystyle\int_{-1}^1 \frac{\ln(x^4+1)}{x^3}\,\mathrm{d}x$

   Answer: 0

2. (a) $\displaystyle\int \frac{1}{(x+1)\sqrt{x}+2x}\,\mathrm{d}x$

   Answer: $-\dfrac{2}{\sqrt{x}+1} + C$

   (b) $\displaystyle\int \frac{1}{(\sqrt{x}+1)x+2\sqrt{x}}\,\mathrm{d}x$

   Answer: $\dfrac{4}{\sqrt{7}}\tan^{-1}\left(\dfrac{2\sqrt{x}+1}{\sqrt{7}}\right) + C$

3. (a) $\displaystyle\int \frac{5x^2+3x-2}{x^3+2x^2}\,\mathrm{d}x$

   Answer: $3\ln|x+2| + 2\ln|x| + \dfrac{1}{x} + C$

   (b) $\displaystyle\int \frac{(x+1)^2}{(x+2)(x^2+x+1)}\,\mathrm{d}x$

   Answer: $\dfrac{1}{3}(\ln(x^2+x+1) + \ln|x+2|) + C$

   Answer: $\dfrac{1}{3}\ln|1+(x+1)^3| + C$

4. (a) $\displaystyle\int_0^\infty \frac{1}{(x+\sqrt{x^2+1})(1+x^2)}\,\mathrm{d}x$

   Answer: $\ln(2)$

   (b) $\displaystyle\int_0^\infty \frac{1}{(x+\sqrt{x^2+1})(1+x)}\,\mathrm{d}x$

   Answer: $-1-\ln(2)-(\sqrt{2}+1)\ln(\sqrt{2}-1)+(\sqrt{2}-1)\ln(\sqrt{2}+1)$

   Answer: $-1-\ln(2)+2\sqrt{2}\ln(1+\sqrt{2})$

   Answer: $-1-\ln(2)+2\sqrt{2}\sinh^{-1}(1)$

   Answer: $-1-\ln(2)+2\sqrt{2}\tanh^{-1}\left(\dfrac{1}{\sqrt{2}}\right)$

5. (a) $\displaystyle\int_0^1 \ln\left(1+\sqrt{1+\left(\frac{x^4-1}{2x^2}\right)^2}\right)\,\mathrm{d}x$

   Answer: $\ln(2)+\pi-2$

   (b) $\displaystyle\int_0^\infty \exp\left(-1-\sqrt{1+\left(\frac{x^4-1}{2x^2}\right)^2}\right)\,\mathrm{d}x$

   Answer: $\dfrac{\sqrt{2\pi}}{2e^2}$

6. (a) $\displaystyle\int_0^\infty \frac{\cos(x)-\cos^2(x)}{x^2}\ln(x)\,\mathrm{d}x$

   Answer: $-\dfrac{\pi\ln(2)}{2}$

   (b) $\displaystyle\int_0^\infty \frac{\cos(x)-\cos^2(x)}{x^2}\,\mathrm{d}x$

   Answer: 0

7. (a) $\displaystyle\lim_{n\to\infty}\int_1^n \ln\left(1+\frac{e}{n}+\frac{1}{\sqrt[e]{nx}}\right)\,\mathrm{d}x$

   Answer: $+\infty$ (accept ∞)

   (b) $\displaystyle\lim_{n\to\infty}\int_1^n \ln\left(1+\frac{2}{n}+\frac{1}{\sqrt{nx}}\right)\,\mathrm{d}x$

   Answer: 4

3 Segment 3

1. (a) $\displaystyle\int_0^1 \frac{x^4+1}{x^6+1}\,\mathrm{d}x$

   Answer: $\dfrac{\pi}{3}$

   (b) $\displaystyle\int_0^1 \frac{x^2+1}{x^3+1}\,\mathrm{d}x$

   Answer: $\dfrac{1}{3}\left(2\ln(2)+\dfrac{\sqrt{3}\pi}{3}\right)$

2. (a) $\displaystyle\int_0^\infty \frac{1}{\sqrt{1+\sqrt{e^x}}}\,\mathrm{d}x$

   Answer: $2(\ln(\sqrt{2}+1)-\ln(\sqrt{2}-1))$

   Answer: $4\sinh^{-1}(1)$

   Answer: $4\tanh^{-1}\left(\dfrac{1}{\sqrt{2}}\right)$

   (b) $\displaystyle\int_1^\infty e^{-\sqrt{\sqrt{x}-1}}\,\mathrm{d}x$

   Answer: 28

3. (a) $\displaystyle\int_0^\infty \frac{\sqrt[3]{x}}{1+x^3}\,\mathrm{d}x$

   Answer: $\dfrac{\pi}{3}\csc\left(\dfrac{4\pi}{9}\right)$

   Answer: $\dfrac{\pi}{3}\sec\left(\dfrac{\pi}{18}\right)$

   (b) $\displaystyle\int_0^\infty \frac{\sqrt[3]{x}}{1+x^4}\,\mathrm{d}x$

   Answer: $\dfrac{\pi}{2\sqrt{3}}$

4. (a) $\displaystyle\int_1^\infty \frac{\tan^{-1}\left(\frac{1}{x^2}\right)}{x}\,\mathrm{d}x$

   Answer: $\dfrac{G}{2}$

   (b) $\displaystyle\int_1^\infty \frac{\tan^{-1}\left(\frac{1}{x}\right)}{x^2}\,\mathrm{d}x$

   Answer: $\dfrac{\pi}{4}-\dfrac{\ln(2)}{2}$

5. (a) $\displaystyle\int_0^\infty \frac{\sqrt[3]{(1+\frac{3}{x})^x}-\sqrt{(1+\frac{2}{x})^x}}{x}\,\mathrm{d}x$

   Answer: $(e-1)\ln\left(\dfrac{2}{3}\right)$

   (b) $\displaystyle\int_{-\infty}^\infty \sin(x^2+x\sqrt{3\pi}+\pi)\,\mathrm{d}x$

   Answer: $\sqrt{\pi}$

6. (a) $\displaystyle\int_0^{\frac{\pi}{4}} \cos^8(x) - \sin^8(x)\,\mathrm{d}x$

   Answer: $\dfrac{5}{12}$

   (b) $\displaystyle\int_0^{\frac{\pi}{4}} \cos^6(x) - \sin^6(x)\,\mathrm{d}x$

   Answer: $\dfrac{11}{24}$

7. (a) $\displaystyle\int_0^1 \sqrt{\frac{x}{1-x}}\ln\left(\frac{x}{1-x}\right)\,\mathrm{d}x$

   Answer: π

   (b) $\displaystyle\int_0^1 \frac{\ln(\sqrt{x}+\sqrt{1-x})}{\sqrt{x}}\,\mathrm{d}x$

   Answer: $2\sqrt{2}\ln(\sqrt{2}+1)-2$